Parameterised CFD Simulation Guide

Hydrogen–Methane T-Junction Mixing
in ANSYS Fluent with Automated Design-Point Sweeps

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Contents

1 Introduction and Problem Definition **4**

1.1 Background 4

1.2 Parametric Variables 4

1.3 Base Operating Conditions 6

1.4 Simulation Strategy Overview 6

2 Parameterised Geometry (DesignModeler) **7**

2.1 Design Rationale 7

2.2 Setup 7

2.3 Main Pipeline 8

2.4 Injection Plane (Fixed per Base) 8

2.5 Parameterised Branch Pipe 9

2.5.1 Branch Extrude Depth Expression 9

2.6 Named Selections (in DesignModeler) 10

2.6.1 Enable Named Selection Transfer in Workbench 10

2.6.2 Creating the Named Selections 10

2.6.3 Post-Update Verification 11

2.7 Material Assignment and Cleanup 12

3 Parameterised Mesh Generation **13**

3.1 Pre-Requisite: Body of Influence in DesignModeler 13

3.1.1 BOI Radius Expression 13

3.2 Global Mesh Settings 14

3.3 Named Selections 14

3.4 Mesh Method 15

3.5 Body-of-Influence Sizing 15

3.6 Inflation Layer (Boundary Layer Mesh) 15

3.6.1 First-Layer Height Derivation 16

3.6.2 Inflation Settings 16

3.7 Mesh Quality Targets 17

4 Fluent Setup — General and Physical Models **18**

4.1 General Settings 18

4.2 Viscous Model (Turbulence) 18

4.3 Species Transport 19
4.4 Energy Equation 20

5 Materials and Operating Conditions 21
5.1 Phase 1 — Import the Pure Gases 21
5.2 Phase 2 — Build the Custom Mixture 21
5.3 Phase 3 — Set Mixture Physical Properties 22
5.4 Individual Species Properties (Verification) 22
5.5 Turbulent Schmidt Number 23
5.6 Operating Conditions 23

6 Boundary Conditions (Parameterised) 24
6.1 Main Inlet (`main_inlet`) — The Natural Gas Crossflow 24
6.2 Branch Inlet (`branch_inlet`) — The Hydrogen Jet 24
6.3 Outlet (`outlet`) — The Downstream Exhaust 25
6.4 Walls (`wall`) — The Pipe Boundaries 26

7 Solution Methods and Controls 27
7.1 Pressure–Velocity Coupling 27
7.2 Spatial Discretisation 27
7.3 Transient Formulation 28
7.4 Under-Relaxation Factors 28

8 Run Setup, Monitors, and Execution 29
8.1 Step 1: Verify Solution Methods (Quick Check) 29
8.2 Step 2: Run Calculation Parameters 29
8.3 Step 3: Residual Monitors 30
8.4 Step 4: Surface Monitors (Track Mixing) 30
8.5 Step 5: Point Monitors (Wall Concentrations) 31
8.6 Step 6: Autosave (Prevents Data Loss) 31
8.7 Step 7: Initialisation 31
8.8 Step 8: Start the Calculation 32
8.9 Step 9: During the Run — What to Watch 32
8.10 Step 10: Enable Time Statistics (After $t = 4\,\mathrm{s}$) 32

9 Workbench Parameterisation and Design Points 33
9.1 Parameter Summary 33
9.2 Defining Expressions 33
9.3 Creating the Design-Point Table 34
9.4 Update Configuration 34

10 Post-Processing and Validation **36**
10.1 Hydrogen Volume Fraction (Custom Field Function) 36
10.2 Key Output Metrics 36
10.2.1 Coefficient of Variation (COV) 36
10.2.2 Wall H2 Concentration 37
10.2.3 Velocity Vectors at Junction 37
10.3 y+ Verification 37
10.4 Mesh Independence Study 38
10.4.1 Three-Mesh Protocol 38
10.4.2 GCI Calculation 38
10.5 Validation Against Reference Data 39

Chapter 1

Introduction and Problem Definition

1.1 Background

Decarbonising domestic heating is a central pillar of net-zero strategies worldwide. A promising transitional approach is to inject **green hydrogen** into the existing natural-gas transmission network at volume fractions of 5–20 %, a range within which most conventional boilers can operate without modification [1].

The injection is typically performed through a **T-junction**—a branch pipe that meets the main pipeline at an angle. Because hydrogen is approximately **eight times lighter** than methane (the dominant component of natural gas), the injected jet lacks momentum to penetrate the main flow and is simultaneously driven upward by buoyancy. The result is *stratification*: localised hydrogen concentrations on the upper pipe wall that can exceed 40 % by volume, far above the bulk target [1]. Prolonged exposure of carbon-steel pipe walls to such concentrations accelerates **hydrogen embrittlement**, reducing fracture toughness and enabling crack propagation [2].

Understanding and controlling the mixing process therefore requires high-fidelity computational fluid dynamics (CFD) that resolves the transient turbulent structures responsible for scalar transport. This guide provides a complete, literature-backed, step-by-step procedure to set up such a simulation in **ANSYS Fluent**, with full parametric automation so that the influence of key geometric and flow variables can be swept systematically.

1.2 Parametric Variables

Three independent parameters govern the T-junction mixing problem. Their definitions, symbols, and proposed sweep ranges are summarised in Table 1.1.

Table 1.1: Parametric variables and their sweep ranges.

Parameter	Symbol	Definition	Sweep Values
Diameter ratio	β	d_2/d_1	0.25, 0.50, 0.75, 1.00
Injection angle	θ	Angle between branch axis and main axis	30°, 60°, 90°, 120°, 150°
Velocity ratio	R_v	u_2/u_1	0.5, 1.0, 2.0, 4.0

To ensure robust parameterisation, the study is organised as **five separate Workbench systems**—one per injection angle. Within each system, only β and R_v are swept as Workbench parameters. This eliminates the topology changes that occur when the injection angle varies (different branch–pipe intersection geometry), guaranteeing that Named Selections, mesh settings, and boundary conditions remain valid across all design points within each base. Table 1.2 summarises the architecture.

Table 1.2: Five-base architecture: one Workbench system per injection angle.

System Name	θ	Design Points	Swept Parameters
`Theta_030`	30°	16	$\beta \times R_v$
`Theta_060`	60°	16	$\beta \times R_v$
`Theta_090`	90°	16	$\beta \times R_v$
`Theta_120`	120°	16	$\beta \times R_v$
`Theta_150`	150°	16	$\beta \times R_v$
Total		**80**	

From these three independent parameters, several derived quantities follow directly:

Volumetric dilution ($\mathcal{D}$):

The ratio of the volumetric flow rate through the branch to that through the main pipe [1]:

$$\mathcal{D} = \beta^2 R_v \tag{1.1}$$

Momentum ratio (M_R):

The ratio of the dynamic pressure of the branch jet to that of the main stream, controlling jet penetration depth [1]:

$$M_R = \frac{\rho_2}{\rho_1} R_v^2 \tag{1.2}$$

Because $\rho_{H_2}/\rho_{CH_4} \approx 1/8$ at equal pressure and similar temperature, a velocity ratio of $R_v = 1$ yields $M_R \approx 0.125$—extremely weak penetration.

1.3 Base Operating Conditions

The simulation replicates the UK high-pressure gas transmission network conditions used by Eames et al. [1]:

Table 1.3: Base operating conditions (fixed across all design points).

Quantity	Value	Source
Main-pipe diameter, d_1	460 mm (18 in)	[1]
Operating pressure, p_{op}	6.9 MPa	[1]
Main-flow velocity, u_1	$10\,\mathrm{m\,s^{-1}}$	[1]
Methane temperature, T_1	284 K	[1]
Hydrogen temperature, T_2	293 K	[1]
Main-flow species	Pure CH_4	[1]
Branch-flow species	Pure H_2	[1]

1.4 Simulation Strategy Overview

The workflow proceeds through five major stages, each detailed in a dedicated chapter. The geometry and meshing stages (Chapters 2 and 3) are repeated **once per injection angle** to create the five base systems. All subsequent stages (Fluent setup, solution, parameterisation) are identical across bases and need only be configured once, then duplicated.

1. **Geometry** (Chapter 2): Build the T-junction fluid domain in DesignModeler with a fixed injection angle and a parameterised branch diameter. Repeat for each of the five angles.

2. **Meshing** (Chapter 3): Generate a mesh with a parameterised Body of Influence and inflation layers targeting $y^+ \approx 1$.

3. **Fluent Physics Setup** (Chapters 4–6): Configure the solver, turbulence model (SA-DDES), species transport, material properties, and boundary conditions.

4. **Solution and Execution** (Chapters 7–8): Set numerical schemes, initialise via Hybrid Initialisation, and run the transient DDES directly from $t = 0$.

5. **Parameterisation and Post-Processing** (Chapters 9–10): Sweep β and R_v within each base system, then combine results across bases for analysis.

Chapter 2

Parameterised Geometry (DesignModeler)

2.1 Design Rationale

The computational domain is a straight circular main pipe with a single circular branch intersecting it. Two key length scales are fixed by the literature:

- **Main-pipe length** $= 20\,d_1 = 9200\,\mathrm{mm}$. A minimum of $10\,d_1$ upstream of the junction is provided as a practical compromise for turbulent profile development at $\mathrm{Re} \sim 10^7$ [3], and $10\,d_1$ downstream provides sufficient mixing length for concentration measurements. Eames et al. showed that stratification persists for at least $8\,d_1$ downstream [1].
- **Branch length** must provide at least 10 branch diameters of straight run for flow development, plus enough extra length to clear the main pipe at the shallowest planned angle.

2.2 Setup

Create one Workbench system per injection angle:

1. In Workbench, drag a **Fluid Flow (Fluent)** system onto the project schematic. Rename it `Theta_090` (or whichever angle you are building first).
2. Repeat for each of the remaining four angles, naming them `Theta_030`, `Theta_060`, `Theta_120`, and `Theta_150`.
3. Double-click the **Geometry** cell of the first system to open DesignModeler.
4. Set **Units** to **Millimeter**.

The steps below describe the geometry construction for a *single* base system. Repeat the entire procedure for each of the five angles, substituting the appropriate value of θ.

2.3 Main Pipeline

1. In the Tree Outline, click **XYPlane** → **New Sketch** (creates `Sketch1`).
2. Switch to the **Sketching** tab → **Draw** → **Circle**. Click the origin and drag outward.
3. **Dimensions** → **General**: click the circle edge and set the diameter to **460**.
4. Switch back to **Modeling**, click **Extrude**:
   - **Operation:** Add Material
   - **Direction:** Normal
   - **Extent Type:** Fixed
   - **Depth: 9200**
5. Click **Generate**.

Result: A 9200 mm cylinder along the Z-axis representing the main pipeline fluid domain.

2.4 Injection Plane (Fixed per Base)

This plane sets the injection angle θ for this base system. Because each angle has its own Workbench system, θ is entered as a **fixed value**—it is *not* parameterised.

1. In the Tree Outline, click **ZXPlane** → **New Plane**.
2. In the Details View, apply:
   - **Transform 1 (RMB): Offset Z** = 4600.
   - **Transform 2 (RMB): Rotate about X** = θ. Enter the fixed angle for this base system (e.g. 90° for `Theta_090`, 30° for `Theta_030`, etc.).
3. **Do not parameterise** this value. Leave the checkbox blank.
4. Click **Generate**.

Why 4600 mm? The branch is placed at $10\,d_1$ from the main inlet, providing sufficient development length for the turbulent profile to establish before reaching the junction.
Why not parameterise θ? Changing the injection angle alters the branch–pipe intersection topology (face count, edge splits), which breaks Named Selections and may cause mesh failures. By fixing θ per base and only sweeping β and R_v, the geometry topology is guaranteed to remain constant across all design points within each base.

2.5 Parameterised Branch Pipe

1. Select the new plane in the tree → **New Sketch** (creates `Sketch2`).
2. **Sketching** → **Draw** → **Circle**. Click the origin of the new plane and draw a circle.
3. **Dimensions** → **General**: set diameter to **115** (the $\beta = 0.25$ baseline, i.e. 0.25×460).
4. **Parameterise** this dimension: click the blank square beside the value. Name the parameter `Branch_Dia`.
5. Switch to **Modeling**, select `Sketch2`, click **Extrude**:
   - **Operation: Add Material**
   - **Direction:** Normal
   - **Depth:** Enter the value from Table 2.1 for this base system's angle.
6. **Parameterise** the depth: click the blank square beside the value. Name the parameter `Branch_Extrude_Depth`.
7. Click **Generate**.

2.5.1 Branch Extrude Depth Expression

Because θ is fixed per base, the clearance term $d_1/(2\sin\theta)$ is a known constant for each system. In the Workbench **Parameter Set**, define `Branch_Extrude_Depth` with the expression:

$$\texttt{Branch_Extrude_Depth} = 10 \times \texttt{Branch_Dia} + \underbrace{\frac{d_1/2}{\sin\theta}}_{\text{constant per base}} \tag{2.1}$$

The clearance constants for each base are pre-computed in Table 2.1.

Table 2.1: Branch extrude depth: fixed clearance term per base system.

Base	θ	$d_1/(2\sin\theta)$	**Workbench Expression**
`Theta_030`	30°	460 mm	`10 * Branch_Dia + 460`
`Theta_060`	60°	266 mm	`10 * Branch_Dia + 266`
`Theta_090`	90°	230 mm	`10 * Branch_Dia + 230`
`Theta_120`	120°	266 mm	`10 * Branch_Dia + 266`
`Theta_150`	150°	460 mm	`10 * Branch_Dia + 460`

This expression contains no trigonometric functions, eliminating the degrees-vs-radians ambiguity that would arise if θ were parameterised.

2.6 Named Selections (in DesignModeler)

Named Selections **must** be created inside DesignModeler rather than in the Meshing application. DesignModeler-based Named Selections are tied to the *generating feature* (e.g. the end face of Extrude 1), so they survive parametric geometry updates when `Branch_Dia` changes.

Because θ is fixed per base system, the branch–pipe intersection topology never changes within a single system—only the branch diameter varies. This guarantees that the face structure (and therefore the Named Selections) remains stable across all 16 design points in each base.

2.6.1 Enable Named Selection Transfer in Workbench

Before proceeding, enable the transfer of Named Selections from DesignModeler to Meshing and Fluent:

1. In Workbench, go to **Tools** → **Options** → **Geometry Import**.
2. Ensure the **Named Selections** checkbox is enabled.
3. Click **OK**.

2.6.2 Creating the Named Selections

Body-Level Named Selections

These are used to scope mesh controls (Method, Sizing, Inflation) and must be created *before* the face-level boundary selections.

1. In DesignModeler, go to **Tools** → **Named Selection**.
2. **`Fluid_Domain`:** Select the `T_Junction_Fluid` body (the main pipe + branch solid). Right-click → **Create Named Selection**. Name it `Fluid_Domain`.
3. **`BOI_Body`:** Select the `BOI_Zone` body (the frozen sphere). Create Named Selection → `BOI_Body`.
4. Click **Generate**.

Face-Level Named Selections (Boundary Conditions)

1. With the complete geometry visible (main pipe + branch + BOI sphere), go to **Tools** → **Named Selection**.
2. **`main_inlet`:** Select the circular face at $Z = 0$ (the flat end of the first extrusion). Right-click → **Create Named Selection**. Name it `main_inlet`.
3. **`outlet`:** Select the circular face at $Z = 9200\,\mathrm{mm}$ (the opposite end of the first extrusion). Create Named Selection → `outlet`.
4. **`branch_inlet`:** Select the flat end face of the branch extrusion (the face farthest from the main pipe). Create Named Selection → `branch_inlet`.
5. **`wall`:** Click **Select All** (to highlight every face), then hold **Ctrl** and click to deselect the three faces already named (`main_inlet`, `outlet`, `branch_inlet`). Create Named Selection → `wall`.
6. Click **Generate**.

2.6.3 Post-Update Verification

After the first design-point update within a base, open the **Mesh** cell and check the **Named Selections** branch in the Outline tree:

- If all four selections display normally, the transfer is intact and the remaining design points will also succeed.
- If any selection shows a **yellow warning icon**, return to DesignModeler, repair the affected Named Selection by re-picking the correct face, and regenerate.

Why this matters: If Named Selections are invalid, boundary conditions in Fluent will map to the wrong faces or disappear entirely. The five-base architecture eliminates the highest-risk topology change (varying θ), but verifying after the first β update within each base is still recommended.

2.7 Material Assignment and Cleanup

1. Expand **1 Part, 1 Body** in the Tree Outline. (The `BOI_Zone` will appear as a second body under a **2 Parts, 2 Bodies** folder.)
2. Right-click the pipe Solid → **Rename** to `T_Junction_Fluid`.
3. Click `T_Junction_Fluid`. In the Details View, change **Fluid/Solid** from `Solid` to **Fluid**.
4. Verify that `BOI_Zone` is also set to **Fluid**.
5. Save the project and close DesignModeler.

Chapter 3

Parameterised Mesh Generation

3.1 Pre-Requisite: Body of Influence in DesignModeler

A Body of Influence (BOI) provides localised mesh refinement in the mixing zone without refining the entire domain. Before entering the Meshing application, create this zone in DesignModeler:

1. **Create** → **Primitives** → **Sphere**.
2. Set **Origin Z** to **4600 mm** (centred on the junction).
3. Set **Operation** to **Add Frozen** (prevents merging with the pipe body).
4. Set the baseline **Radius** to **690 mm** ($= 1.5\,d_1$).
5. **Parameterise** the radius: click the blank square beside the value and name the parameter `BOI_Radius`.
6. Click **Generate**. Rename the new body to `BOI_Zone` and set its material to **Fluid**.

3.1.1 BOI Radius Expression

In the Parameter Set, define:

$$\texttt{BOI_Radius} = \max\Big(1.5\,d_1,\ \ 8 \times \texttt{Branch_Dia}\Big) \tag{3.1}$$

Justification: Eames et al. [1] demonstrated that hydrogen stratification persists for up to 8 main-pipe diameters downstream of the injection point. The refinement sphere must enclose this region. The max function ensures adequate coverage for both small and large branch diameters.

Workbench expression syntax: If `Branch_Dia` is parameter `P2` and $d_1 = 460\,\text{mm}$, type:
`max(690, P2 * 8)`

3.2 Global Mesh Settings

1. In Workbench, double-click the **Mesh** cell.
2. Click **Mesh** in the Outline tree. In the Details panel:
   - **Physics Preference:** CFD
   - **Solver Preference:** Fluent
3. Under **Sizing**:
   - **Element Size:** 15 mm ($\approx d_1/30$; provides a coarse far-field mesh that keeps cell count manageable)
   - **Growth Rate:** 1.15 (industry standard for smooth transitions [4])
   - **Capture Curvature:** Yes
   - **Curvature Normal Angle:** 15° (ensures circular cross-sections are geometrically faithful)
4. Under **Advanced**: set **Element Order** to **Linear**.

3.3 Named Selections

Six Named Selections were created in DesignModeler (Section 2.6): two body-level (`Fluid_Domain`, `BOI_Body`) and four face-level (`main_inlet`, `branch_inlet`, `outlet`, `wall`). Verify that all six appear in the **Named Selections** branch of the Meshing Outline tree. Do **not** redefine them here—doing so would create fragile, ID-based selections that break during parametric updates.

Why body-level Named Selections matter: All mesh controls below (Method, BOI Sizing, Inflation) are scoped to `Fluid_Domain` or `BOI_Body` via Named Selections instead of direct geometry picking. Direct picking (e.g. Ctrl+B on a body) creates a hard-coded reference to an internal body ID. When `Branch_Dia` changes and the geometry regenerates, these IDs can shift, causing the mesh control to lose its scoping attachment and silently fail. Named-Selection-based scoping is resolved by *name*, making it immune to ID renumbering.

3.4 Mesh Method

1. Right-click **Mesh** → **Insert** → **Method**.
2. **Scoping Method: Named Selection**.
3. **Named Selection:** `Fluid_Domain`.
4. **Method: MultiZone**.
5. **Src/Trg Selection:** Automatic.
6. **Free Mesh Type: Hexa Core**.

Mesh method selection per base: Because θ is fixed per base, you can choose the optimal mesh method for each angle independently. MultiZone with Hexa Core works well for $\theta \geq 60°$. For the `Theta_030` base (shallow intersection), MultiZone may fail due to the complex junction topology. In that case, switch the Method to **Tetrahedrons** and convert to polyhedra inside Fluent (`Mesh` → `Polyhedra` → `Convert`). Polyhedral cells reduce cell count by ∼4× while maintaining accuracy [4].

3.5 Body-of-Influence Sizing

1. Right-click **Mesh** → **Insert** → **Sizing**.
2. **Scoping Method: Named Selection**.
3. **Named Selection:** `Fluid_Domain`.
4. **Type: Body of Influence**.
5. **Body of Influence:** Change **Scoping Method** to **Named Selection**, then select `BOI_Body`.
6. **Element Size: 5 mm**.

Justification: For the baseline case ($\beta = 0.25$, $d_2 = 115\,\text{mm}$), this gives ∼23 cells across the branch diameter, sufficient to resolve the jet shear layer. Gritskevich et al. [5] demonstrated adequate resolution of T-junction mixing with comparable cell-per-diameter ratios in their OECD benchmark.

3.6 Inflation Layer (Boundary Layer Mesh)

A resolved boundary layer ($y^+ \approx 1$) is mandatory for the Spalart–Allmaras DDES model, which does not use wall functions [6, 7].

3.6.1 First-Layer Height Derivation

The first cell height Δy_1 for a target $y^+ = 1$ is derived as follows [3]:

1. **Reynolds number** based on main-pipe diameter and methane properties at $p = 6.9\,\mathrm{MPa}$, $T = 284\,\mathrm{K}$:

$$\rho_1 = \frac{M_{\mathrm{CH_4}}\, p}{R_g\, T_1} = \frac{0.01604 \times 6.9 \times 10^6}{8.314 \times 284} \approx 46.9\ \mathrm{kg\,m^{-3}} \tag{3.2}$$

$$\mu_1 \approx 1.25 \times 10^{-5}\ \mathrm{Pa\,s} \qquad \text{(NIST [8] at 6.9 MPa, 284 K)} \tag{3.3}$$

$$\mathrm{Re}_{d_1} = \frac{\rho_1\, u_1\, d_1}{\mu_1} = \frac{46.9 \times 10 \times 0.46}{1.25 \times 10^{-5}} \approx 1.73 \times 10^7 \tag{3.4}$$

2. **Skin-friction coefficient** via the Schlichting correlation for turbulent pipe flow [3]:

$$C_f = 0.058\ \mathrm{Re}_{d_1}^{-0.2} = 0.058 \times (1.73 \times 10^7)^{-0.2} \approx 2.11 \times 10^{-3} \tag{3.5}$$

3. **Wall shear stress** and **friction velocity**:

$$\tau_w = \tfrac{1}{2}\, C_f\, \rho_1\, u_1^2 = \tfrac{1}{2} \times 2.11 \times 10^{-3} \times 46.9 \times 100 \approx 4.95\ \mathrm{Pa} \tag{3.6}$$

$$u_\tau = \sqrt{\tau_w/\rho_1} = \sqrt{4.95/46.9} \approx 0.325\ \mathrm{m\,s^{-1}} \tag{3.7}$$

4. **First cell height** for $y^+ = 1$:

$$\Delta y_1 = \frac{y^+\, \mu_1}{\rho_1\, u_\tau} = \frac{1 \times 1.25 \times 10^{-5}}{46.9 \times 0.325} \approx 8.2 \times 10^{-7}\ \mathrm{m} \approx \boxed{0.001\ \mathrm{mm}} \tag{3.8}$$

Result: At 6.9 MPa operating pressure, a first-layer height of **0.001 mm (1 μm)** achieves $y^+ \approx 1$ for the methane main flow. The high gas density at this pressure ($\sim$47 kg m^{-3} versus $\sim$0.66 kg m^{-3} at atmospheric) is the dominant factor requiring such a thin first cell. This value should be verified against the post-processed y^+ field after the first simulation run.

3.6.2 Inflation Settings

1. Right-click **Mesh** → **Insert** → **Inflation**.
2. **Scoping Method: Named Selection**.
3. **Named Selection:** `Fluid_Domain`.

4. **Boundary Scoping Method: Named Selections**.
5. **Boundary:** Select the `wall` named selection.
6. **Inflation Option: First Layer Thickness**.
7. **First Layer Height: 0.001 mm**.
8. **Maximum Layers: 20**.
9. **Growth Rate: 1.2**.

Layer count justification: With 20 layers and a growth rate of 1.2, the inflation stack spans approximately $\Delta y_1 \times (1.2^{20} - 1)/(1.2 - 1) \approx 0.001 \times 186.7 \approx 0.19\,\text{mm}$ from the wall. The turbulent boundary-layer thickness δ for the main pipe can be estimated as $\delta \approx 0.37\, d_1\, \text{Re}_{d_1}^{-0.2} \approx 0.37 \times 460 \times (1.73 \times 10^7)^{-0.2} \approx 6.2\,\text{mm}$ [3]. The inflation stack covers the viscous sublayer and buffer layer; the outer boundary layer is resolved by the BOI cells (5 mm).

3.7 Mesh Quality Targets

After generating the mesh (**Mesh** → **Generate**), verify the following quality metrics in the **Statistics** panel:

Table 3.1: Mesh quality acceptance criteria.

Metric	Target	Reference
Minimum Orthogonal Quality	> 0.10	[4]
Maximum Skewness	< 0.95	[4]
Maximum Aspect Ratio	< 100	[4]

Chapter 4

Fluent Setup — General and Physical Models

Double-click the **Setup** cell in Workbench to launch Fluent. When prompted, select **Double Precision** and set the number of **Solver Processes** to 8 (appropriate for a $\sim$3 M cell mesh on an i7-14700 with 8 P-cores).

4.1 General Settings

Navigate to **Setup** $\rightarrow$ **General**:

1. **Type:** Pressure-Based. (Mach ≈ 0.02)
2. **Velocity Formulation:** Absolute.
3. **Time: Transient**. (SA-DDES is inherently unsteady; no steady RANS precursor.)
4. **Gravity: Enabled**. Set $g_x = 0$, $g_y = -9.81$, $g_z = 0$. (Buoyancy drives hydrogen stratification; density ratio CH_4:H_2 ≈ 8 at 6.9 MPa.)

4.2 Viscous Model (Turbulence)

Navigate to **Models** $\rightarrow$ **Viscous**:

1. **Model: Spalart–Allmaras (1 eqn)**.
2. Under **Spalart–Allmaras Options**:
   - **Production:** Change from Vorticity-Based (default) to **Strain/Vorticity-Based**.
   - **Curvature Correction: Enable**.
3. In the left list, select **Detached Eddy Simulation (DES)**.

4. **DES Option: Delayed DES (DDES)** (default).
5. **Model Constants:** Leave at defaults (well-validated).
6. Click **OK**.

Rationale:

- SA-DDES chosen by Eames et al. [1]; DDES shielding prevents premature LES activation in the boundary layer [7, 9].
- **Strain/Vorticity-Based Production** reduces over-prediction of eddy viscosity in high-shear regions such as the hydrogen jet mixing zone. This is a known improvement for free shear flows [6].
- **Curvature Correction** prevents the SA model from artificially damping resolved vortices in the pipe curvature and swirling wake downstream of the junction [7].

Visual confirmation: After applying settings, the Viscous Model dialog should display:	
Model:	Detached Eddy Simulation (DES)
RANS Model:	Spalart–Allmaras
DES Option:	Delayed DES (DDES)
Production:	Strain/Vorticity-Based
Curvature Correction:	Enabled

4.3 Species Transport

Navigate to **Models** → **Species**:

1. Select **Species Transport**.
2. **Reactions:** Disabled. (Mechanical mixing study, not combustion; saves significant computation.)
3. **Diffusion Energy Source:** Enabled. (H_2 and CH_4 have very different c_p and thermal conductivities; as they diffuse into each other, enthalpy is transported.)
4. **Full Multicomponent Diffusion:** Disabled. (With only two species—H_2 and CH_4—Fickian binary diffusion is exact [10]. Enable only if modelling natural gas as a multi-component blend.)

4.4 Energy Equation

Enable the Energy Equation. At 6.9 MPa the gas is highly compressible; the solver needs the energy equation to recalculate density via the ideal gas law ($\rho = p\,M/R_g\,T$) as pressure and mixing ratios fluctuate in the T-junction [11].

Chapter 5

Materials and Operating Conditions

5.1 Phase 1 — Import the Pure Gases

Fluent has separate databases for **Mixtures** and pure **Fluids**. The pure species must be copied into your project before a custom mixture can reference them.

1. In the Fluent tree, expand **Materials**.
2. Double-click **Fluid** (default entry is `air`). The *Create/Edit Materials* window opens.
3. Click **Fluent Database...** on the right.
4. In the database window, ensure **Material Type** is set to **`fluid`** (not `mixture`—selecting `mixture` shows pre-built templates like `methane-air`, which is not what we want).
5. Scroll to **methane (ch4)**, click it, then click **Copy**.
6. Scroll to **hydrogen (h2)**, click it, then click **Copy**.
7. Click **Close**, then close the *Create/Edit Materials* window.

5.2 Phase 2 — Build the Custom Mixture

1. Under **Materials**, double-click **Mixture** (`mixture-template`).
2. In the row **Mixture Species**, click **Edit...**.
3. The *Species* dialog opens with *Available Materials* (left) and *Selected Species* (right).
4. Add **ch4** and **h2** from left to right.
5. Remove **o2** and **n2** from the right list.
6. **Order matters:** the last species in the right list is the "balance" species (computed from $1 - \sum Y_i$). Since the pipe is mostly CH_4, ensure the list reads:

(a) `h2`

(b) `ch4` (bottom = balance species)

7. Click **OK**.

5.3 Phase 3 — Set Mixture Physical Properties

Back on the *Create/Edit Materials* window for the mixture, set each property drop-down:

Table 5.1: Mixture-level property settings.

Property	**Setting**
Density	**ideal-gas**
c_p (Specific Heat)	**mixing-law**
Thermal Conductivity	**ideal-gas-mixing-law**
Viscosity	**ideal-gas-mixing-law**
Mass Diffusivity	**kinetic-theory**

Click **Change/Create**, then **Close**.

Note: Ideal-gas is a modelling simplification (Eames et al. used it [1]). Using mixing-law and ideal-gas-mixing-law for c_p, conductivity, and viscosity lets Fluent weight each property by local species concentration automatically. Temperature-dependent polynomials are negligible for $\Delta T < 10\,\mathrm{K}$ [10].

5.4 Individual Species Properties (Verification)

After the mixture is created, verify the per-species constants match NIST data [8] at 6.9 MPa:

Table 5.2: Species thermophysical properties.

Property	**CH_4**	**H_2**	**Source**
Molecular weight ($\mathrm{kg\,kmol^{-1}}$)	16.043	2.016	NIST
c_p ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	2580	14 500	NIST
Thermal conductivity ($\mathrm{W\,m^{-1}\,K^{-1}}$)	0.038	0.185	NIST
Dynamic viscosity ($\mathrm{Pa\,s}$)	1.25×10^{-5}	9.0×10^{-6}	NIST

These values are loaded from the Fluent database when you copy the pure fluids. If any differ significantly from the table above, edit them manually in the species material panel.

5.5 Turbulent Schmidt Number

Default $\mathrm{Sc}_t = 0.7$; for H_2 jets in crossflow, $\mathrm{Sc}_t = 0.5$ improves mixing prediction. Change via TUI:

```
/define/models/viscous/turbulent-schmidt-number 0.5
```

5.6 Operating Conditions

Navigate to **Cell Zone Conditions** → **Operating Conditions**:

1. **Operating Pressure:** Enter **6900000** (6.9 MPa; do not use commas).
2. **Reference Pressure Location:** Leave X, Y, Z at default (0, 0, 0).
3. **Gravity:** Check the **Gravity** box. Set $g_y = -9.81\,\mathrm{m\,s^{-2}}$ (branch points along Y after the 90° ZX-plane rotation); leave $g_x = g_z = 0$.
4. **Specified Operating Density:** Check the box, enter **0**.

Critical: Incorrect operating pressure changes density by two orders of magnitude, invalidating all physics.

Why Operating Density = 0? With gravity enabled, Fluent computes buoyancy as $(\rho - \rho_0)\,\mathbf{g}$. For a compressible, multi-species mixture (ideal gas with variable composition), the local density ρ varies significantly with pressure, temperature, and species concentration. A non-zero ρ_0 introduces an artificial hydrostatic baseline that can destabilise the calculation. Setting $\rho_0 = 0$ reduces the buoyancy term to $\rho\,\mathbf{g}$, using the true local density — the correct formulation for compressible flows [11].

Chapter 6

Boundary Conditions (Parameterised)

Expand **Boundary Conditions** in the Fluent tree. All named selections from meshing appear automatically.

6.1 Main Inlet (`main_inlet`) — The Natural Gas Cross-flow

Double-click `main_inlet`. Ensure **Type** is `velocity-inlet`.

1. **Momentum Tab:**
   - Velocity Magnitude: **10** (m/s).
   - Turbulence Specification Method: **Intensity and Hydraulic Diameter**.
   - Turbulent Intensity: **5** (%).
   - Hydraulic Diameter: **0.46** (m).
2. **Thermal Tab:**
   - Temperature: **284** (K).
3. **Species Tab:**
   - Species Mass Fractions (h2): **0**. (Pure methane.)
4. Click **Apply** and close.

6.2 Branch Inlet (`branch_inlet`) — The Hydrogen Jet

Double-click `branch_inlet`. Ensure **Type** is `velocity-inlet`.

1. **Momentum Tab:**

- Velocity Magnitude: **32** (m/s).
- **Parameterise:** Click the small drop-down arrow (or blank square) to the right of the velocity value box → select **New Input Parameter...** → name it `branch_velocity` → click OK. A blue "P" appears next to the box, linking it to the Workbench parameter table (Chapter 9).
- Turbulence Specification Method: **Intensity and Hydraulic Diameter**.
- Turbulent Intensity: **5** (%).
- Hydraulic Diameter: **0.115** (m).

2. **Thermal Tab:**
   - Temperature: **293** (K).
3. **Species Tab:**
   - Species Mass Fractions (h2): **1**. (100 % pure hydrogen.)
4. Click **Apply** and close.

Velocity derivation: $\mathcal{D} = \beta^2(u_2/u_1) \;\Rightarrow\; 0.2 = (0.25)^2(u_2/10) \;\Rightarrow\; u_2 = 32\,\mathrm{m\,s^{-1}}$. In the Parameter Set, use `branch_velocity` $= R_v \times 10$ to sweep velocity ratio.

6.3 Outlet (`outlet`) — The Downstream Exhaust

Double-click `outlet`. Ensure **Type** is `pressure-outlet`.

1. **Momentum Tab:**
   - Gauge Pressure: **0**. (0 gauge = 6.9 MPa absolute, since operating pressure is already set.)
   - Backflow Turbulence Specification: **Intensity and Hydraulic Diameter**.
   - Backflow Turbulent Intensity: **5** (%).
   - Backflow Hydraulic Diameter: **0.46** (m).
2. **Thermal Tab:**
   - Backflow Total Temperature: **284** (K).
3. **Species Tab:**
   - Backflow Species Mass Fractions (h2): **0**.

4. Click **Apply** and close.

Backflow parameters are safety nets Fluent uses if the swirling vortex momentarily pulls fluid backwards through the exit.

6.4 Walls (`wall`) — The Pipe Boundaries

Double-click `wall`. Ensure **Type** is `wall`.

1. **Momentum Tab:** Verify **Stationary Wall** and **No Slip** (defaults).
2. **Thermal Tab:** Set thermal condition to **Heat Flux** with value **0** $\mathrm{W\,m^{-2}}$ (adiabatic — buried pipe, no heat exchange [1]).
3. Click **Apply** and close.

Chapter 7

Solution Methods and Controls

7.1 Pressure–Velocity Coupling

Navigate to **Solution** → **Methods**.

1. Under **Pressure-Velocity Coupling**, click the **Scheme** drop-down.
2. Select **PISO**.
3. Two options appear below (Skewness Correction, Neighbor Correction) — leave at defaults (usually 1).

7.2 Spatial Discretisation

Still in **Solution** → **Methods**, set each drop-down in the **Spatial Discretization** section:

1. **Gradient:** `Least Squares Cell Based` (default).
2. **Pressure:** Change to **`PRESTO!`** (crucial for the buoyancy/gravity settings).
3. **Momentum:** Change to **`Bounded Central Differencing`** (the most important setting for DDES).
4. **Modified Turbulent Viscosity:** Change to **`Second Order Upwind`**.
5. **Species:** Change to **`Second Order Upwind`**.
6. **Energy:** Change to **`Second Order Upwind`**.

Critical: Standard Upwind schemes for momentum create numerical friction that instantly kills the turbulent swirling eddies DDES is designed to resolve. Bounded Central Differencing is the default DES setting in Fluent and must not be changed [11, 12].

7.3 Transient Formulation

Still in **Solution** → **Methods**, at the bottom of the panel:

1. Under **Transient Formulation**, click the drop-down.
2. Select **`Bounded Second Order Implicit`**.

Prevents oscillations while maintaining temporal accuracy [12].

7.4 Under-Relaxation Factors

Navigate to **Solution** → **Controls**. These act as mathematical "brakes" preventing divergence if pressure changes too violently between time steps. Match the following values:

Table 7.1: Under-relaxation factors (transient PISO).

Variable	**Factor**
Pressure	0.3
Momentum	0.7
Modified Turbulent Viscosity	0.8
Species	0.9
Energy	1.0

If **Density** or **Body Forces** appear in the list, leave them at their default value (1.0).

Chapter 8

Run Setup, Monitors, and Execution

Follow these steps **in order** after setting the under-relaxation factors. Do not skip any.

8.1 Step 1: Verify Solution Methods (Quick Check)

Go to **Solution** → **Methods** and confirm:

- Pressure–Velocity Coupling: `PISO`
- Transient Formulation: `Bounded Second Order Implicit`
- Momentum: `Bounded Central Differencing` (*not* Upwind)
- Pressure: `PRESTO!`
- All others: `Second Order Upwind`

If Momentum is `Second Order Upwind`, change it now — otherwise the DES will not resolve turbulence.

8.2 Step 2: Run Calculation Parameters

Go to **Solution** → **Run Calculation** and enter:

Table 8.1: Run calculation settings.

Field	Value	Explanation
Time Step Size (s)	`2e-4`	$\Delta t = 2 \times 10^{-4}$ s; CFL ≈ 1.3
Number of Time Steps	`25000`	$0.0002 \times 25{,}000 = 5$ s total
Max Iterations / Time Step	`20`	Enough for residuals to drop 1–2 orders

Do not click Calculate yet — first set up monitors and autosave.

8.3 Step 3: Residual Monitors

1. **Solution** → **Monitors** → **Residuals** → **Edit**.
2. Check all equations: continuity, x-velocity, y-velocity, z-velocity, modified turbulent viscosity (`nut`), species (`h2`), energy.
3. **Convergence Criterion:** `1e-4` for all; set energy to `1e-6`.
4. Ensure **Plot** is enabled → click **OK**.

8.4 Step 4: Surface Monitors (Track Mixing)

Right-click **Monitors** → **New** → **Surface Monitor...** for each of the following:

Monitor 1 — Outlet H_2 mass fraction:

- Name: `outlet_h2`
- Report Type: `Mass-Weighted Average`
- Field Variable: Species → Mass Fraction of h2
- Surfaces: `outlet`
- Click **Create**.

Monitor 2 — Outlet mass flow (mass balance):

- Name: `outlet_massflow`
- Report Type: `Mass Flow Rate`
- Surfaces: `outlet`
- Click **Create**.

Monitor 3 — Main inlet mass flow:

- Name: `main_inlet_massflow`
- Report Type: `Mass Flow Rate`
- Surfaces: `main_inlet`
- Click **Create**.

(Repeat for `branch_inlet` if desired.)

8.5 Step 5: Point Monitors (Wall Concentrations)

Track H_2 mass fraction at the pipe-top wall ($y = +0.23\,\mathrm{m}$) at four downstream locations. Right-click **Monitors** → **New** → **Point Monitor...** for each row:

Table 8.2: Point monitor coordinates (junction at $Z = 4.6\,\mathrm{m}$, $d_1 = 0.46\,\mathrm{m}$).

Name	X (m)	Y (m)	Z (m)	**Position**
`wall_h2_z2d`	0	0.23	5.52	$z/d_1 = 2$
`wall_h2_z4d`	0	0.23	6.44	$z/d_1 = 4$
`wall_h2_z6d`	0	0.23	7.36	$z/d_1 = 6$
`wall_h2_z8d`	0	0.23	8.28	$z/d_1 = 8$

For each: **Variable** → Species → Mass Fraction of h2 → click **Create**.

8.6 Step 6: Autosave (Prevents Data Loss)

1. Go to **Solution** → **Run Calculation** → click **Activities** (bottom of panel).
2. Click **Create** → **Autosave**.
3. Save Data File Every (Time Steps): **500**.
4. Save Case File Every: **500**.
5. *Retain Only the Most Recent Files:* Optional (uncheck to keep all).
6. Click **OK**, then close Activities.

This saves every 0.1 s ($500 \times 2 \times 10^{-4}$). Adjust if disk space is limited.

8.7 Step 7: Initialisation

1. Navigate to **Solution** → **Initialization**.
2. Select **Hybrid Initialization** → click **Initialize**.
3. **Patch branch region:**
   - Click **Patch...**.
   - Create a cell register (sphere/cylinder around the branch junction).
   - Set H_2 mass fraction = 1.0 and velocity = $32\,\mathrm{m\,s^{-1}}$ along the branch axis.

8.8 Step 8: Start the Calculation

Return to **Solution** → **Run Calculation** → click **Calculate**.

Residuals and monitor plots will update in real time.

8.9 Step 9: During the Run — What to Watch

Table 8.3: Runtime diagnostics.

Observation	**Action**
Residuals drop 1–2 orders each time step	Normal
Residuals stagnate or increase	Reduce Δt to `1e-4` for a few hundred steps
Outlet H_2 mass fraction → ∼0.149 ($\mathcal{D} = 0.2$)	Good
Mass flow imbalance > 1 %	Check boundary conditions
Temperature goes negative	Reduce Δt; check energy equation

8.10 Step 10: Enable Time Statistics (After $t = 4\,\mathrm{s}$)

After the first 20 000 time steps ($t = 4\,\mathrm{s}$), the initial transients have washed out. Enable time-averaged statistics for the final 1 s:

1. **Solution** → **Run Calculation** → under **Data Sampling**, check **Enable**.
2. **Sampling Interval:** `1` (every time step).
3. Let the simulation continue to $t = 5\,\mathrm{s}$ (step 25 000).

Summary checklist:

- Under-relaxation factors set ☐
- Solution methods verified (BCD for momentum) ☐
- Run parameters entered ($\Delta t = 2 \times 10^{-4}$, 25 000 steps, 20 iter) ☐
- Residual monitors enabled ☐
- Surface monitors created (outlet H_2, mass flows) ☐
- Point monitors created (wall H_2 at $2d$, $4d$, $6d$, $8d$) ☐
- Autosave configured (every 500 steps) ☐
- Initialisation complete (Hybrid + Patch) ☐
- Clicked **Calculate** ☐

Chapter 9

Workbench Parameterisation and Design Points

9.1 Parameter Summary

Each of the five base systems contains the same set of Workbench parameters. After completing the Geometry, Mesh, and Fluent setup for a single base, the following parameters should be visible when you double-click the **Parameter Set** bar:

Table 9.1: Workbench parameters (identical in each of the five bases).

Parameter Name	Source	Type	Baseline
`Branch_Dia`	DesignModeler (Sketch dim.)	Input	115 mm
`Branch_Extrude_Depth`	DesignModeler (Extrude depth)	Output	(expression)
`BOI_Radius`	DesignModeler (Sphere radius)	Output	(expression)
`branch_velocity`	Fluent (BC velocity mag.)	Input	$32\,\mathrm{m\,s^{-1}}$

`Branch_Extrude_Depth` and `BOI_Radius` are *output* parameters driven by expressions linked to `Branch_Dia`. They update automatically. The injection angle θ is **not** a Workbench parameter—it is baked into each base system's geometry.

9.2 Defining Expressions

In the Parameter Set table, click on the **Expression** cell for each driven parameter and enter:

1. `Branch_Extrude_Depth`: Enter the expression from Table 2.1 corresponding to this base's angle.

   Example for `Theta_090:` `10 * Branch_Dia + 230`

2. `BOI_Radius`: `max(690, Branch_Dia * 8)`

(690 mm = 1.5 × d_1; identical across all bases.)

9.3 Creating the Design-Point Table

Within each base system, the design-point table sweeps `Branch_Dia` (β) and `branch_velocity` (R_v). With 4 levels of each, the full factorial yields $4 \times 4 = 16$ design points per base.

1. In the Parameter Set, click **Design Points** (bottom panel).
2. Populate the table using the mapping:

$$\texttt{Branch_Dia} = \beta \times 460$$
$$\texttt{branch_velocity} = R_v \times 10$$

3. The complete 16-point table for each base is shown in Table 9.2.

Table 9.2: Design points within each base system (16 per base, 80 total).

DP	β	R_v	`Branch_Dia`	`branch_velocity`
1	0.25	0.5	115 mm	5 m/s
2	0.25	1.0	115 mm	10 m/s
3	0.25	2.0	115 mm	20 m/s
4	0.25	4.0	115 mm	40 m/s
5	0.50	0.5	230 mm	5 m/s
6	0.50	1.0	230 mm	10 m/s
7	0.50	2.0	230 mm	20 m/s
8	0.50	4.0	230 mm	40 m/s
9	0.75	0.5	345 mm	5 m/s
10	0.75	1.0	345 mm	10 m/s
11	0.75	2.0	345 mm	20 m/s
12	0.75	4.0	345 mm	40 m/s
13	1.00	0.5	460 mm	5 m/s
14	1.00	1.0	460 mm	10 m/s
15	1.00	2.0	460 mm	20 m/s
16	1.00	4.0	460 mm	40 m/s

9.4 Update Configuration

For each base system:

1. Right-click the **Parameter Set** bar → **Properties**.
2. Set **Design Point Update** to **Update Sequentially** (most stable for transient DDES).
3. Set **Update Action** to **Update Full Project** (ensures geometry, mesh, and solver all update for each design point).
4. Click **Update All Design Points** to begin the automated sweep for this base.
5. After the 16 design points complete, move to the next base system and repeat.

Computation time estimate: Each transient DDES simulation of 25 000 time steps with 20 iterations each requires ∼500 000 solver iterations. On an 8-core workstation, expect ∼16–32 hours per design point (depending on mesh size). Each base (16 points) requires 256–512 core-hours. The full study across all five bases (80 design points) requires 1280–2560 core-hours.
Parallelisation tip: Because the five base systems are independent Workbench systems, they can run on separate machines simultaneously, reducing wall-clock time by up to 5×.

Chapter 10

Post-Processing and Validation

10.1 Hydrogen Volume Fraction (Custom Field Function)

The reference paper [1] reports hydrogen concentration as **volume fraction** ϕ_{H_2}, while Fluent solves for **mass fraction** Y_{H_2}. To enable direct comparison, define a custom field function for the conversion:

$$\phi_{\mathrm{H}_2} = \frac{Y_{\mathrm{H}_2}/M_{\mathrm{H}_2}}{Y_{\mathrm{H}_2}/M_{\mathrm{H}_2} + (1 - Y_{\mathrm{H}_2})/M_{\mathrm{CH}_4}} \tag{10.1}$$

In Fluent, navigate to **Custom Field Functions** → **Define** and enter:

```
(Y_H2/2.016) / (Y_H2/2.016 + (1-Y_H2)/16.043)
```

Name it `vol_frac_H2`. Use this function for all contour plots, line plots, and cross-sectional averages when comparing against the reference data.

10.2 Key Output Metrics

For each design point, extract and record the following quantities from the time-averaged fields:

10.2.1 Coefficient of Variation (COV)

The COV quantifies mixing uniformity on a cross-sectional plane [14]:

$$\mathrm{COV} = \frac{\sigma_Y}{\bar{Y}} \tag{10.2}$$

where $\bar{Y}$ is the area-weighted mean H_2 mass fraction on the plane and σ_Y is the standard deviation. A fully mixed flow has COV = 0.

In Fluent, compute the COV using:

1. Navigate to **Results** → **Surface Integrals**.
2. Select the downstream iso-surface (e.g. $Z = 9200\,\text{mm}$).
3. Report both **Area-Weighted Average** and **Standard Deviation** of the H2 mass fraction.
4. Compute COV = Std. Dev./Mean.

10.2.2 Wall H_2 Concentration

To assess embrittlement risk, visualise the time-averaged H_2 mass fraction on the pipe wall:

1. **Results** → **Contours**.
2. **Surface:** `wall`.
3. **Field Variable:** Species → Mean Mass Fraction of h2 (from time statistics).
4. **Colormap range:** 0 to 0.05 (to highlight regions exceeding 5% mass fraction).

Record the **maximum** H_2 mass fraction on the wall and its axial location downstream of the junction. Eames et al. [1] reported wall concentrations exceeding 40% by volume for top-side injection at $\beta = 0.25$—an important validation benchmark.

10.2.3 Velocity Vectors at Junction

1. Create a **Plane** at $Z = 4600\,\text{mm}$ (through the junction centre).
2. **Results** → **Vectors**.
3. **Surface:** The plane created above.
4. Colour by **H2 mass fraction** to visualise both the flow pattern and species distribution simultaneously.

10.3 y^+ Verification

After the first simulation completes:

1. **Results** → **Contours**.
2. **Surface:** `wall`.

3. **Field Variable:** Wall Fluxes → Wall Yplus (or y^+).

Verify that $y^+ \leq 1$ over the majority of the wall surface. Localised regions near the junction stagnation point may have slightly elevated values ($y^+ \sim 2$–3), which is acceptable. If y^+ consistently exceeds 5, the first-layer height must be reduced and the mesh regenerated.

10.4 Mesh Independence Study

A mesh independence study must be performed for the baseline design point ($\beta = 0.25$, $R_v = 1.0$) in **at least one** base system (recommended: `Theta_090`) before committing to the full parametric sweep. If the mesh topology differs between bases (e.g. tetrahedra for `Theta_030` vs. MultiZone for `Theta_090`), perform a separate GCI study for each distinct mesh method.

Follow the Grid Convergence Index (GCI) procedure of Roache [15] and Celik et al. [16]:

10.4.1 Three-Mesh Protocol

1. Generate three systematically refined meshes by scaling the global element size and BOI element size:

Table 10.1: Mesh refinement levels for GCI study.

Mesh	Global Size	BOI Size	Refinement Ratio (r)
Coarse	22.5 mm	7.5 mm	—
Medium (baseline)	15 mm	5 mm	1.5
Fine	10 mm	3.3 mm	1.5

2. Run each mesh through the complete transient DDES workflow (Hybrid Initialisation → transient run).

3. Extract the same output quantities (COV at outlet, maximum wall H_2 concentration, pressure drop) from each mesh.

10.4.2 GCI Calculation

For a monitored quantity ϕ from the three meshes (ϕ_1 fine, ϕ_2 medium, ϕ_3 coarse):

1. Compute the observed order of convergence:

$$p = \frac{\ln\left|\frac{\phi_3-\phi_2}{\phi_2-\phi_1}\right|}{\ln(r)} \tag{10.3}$$

2. Compute the fine-grid GCI:

$$\mathrm{GCI}_{\mathrm{fine}} = \frac{F_s \left|\frac{\phi_2-\phi_1}{\phi_1}\right|}{r^p - 1} \tag{10.4}$$

   where $F_s = 1.25$ is the safety factor for three or more grids [15].

3. A GCI $< 5\,\%$ indicates that the medium mesh is sufficiently resolved for the parametric sweep.

10.5 Validation Against Reference Data

For the baseline case ($\beta = 0.25$, R_v corresponding to $\mathcal{D} = 0.2$) in the `Theta_090` base, compare results against Eames et al. [1]:

- **Qualitative:** H_2 concentration contours on the pipe centreline plane should show the characteristic stratified plume for top-side injection and improved mixing for alternative injection orientations.
- **Quantitative:** The area-weighted H_2 mass fraction at $5\,d_1$ and $10\,d_1$ downstream should be within $10\,\%$ of the reference values.
- **Maximum wall concentration:** Should reproduce the $>40\,\%$ wall concentration reported for top-side injection at low dilution ratios.

Any systematic discrepancy may arise from differences between the OpenFOAM solver (used by Eames et al.) and ANSYS Fluent, particularly in the DES shielding function implementation. Such differences should be documented and discussed.

Cross-base comparison: After completing all five bases, combine the results into a single analysis matrix. Plot COV and maximum wall H_2 concentration as functions of θ, β, and R_v to identify trends and interactions. This three-dimensional parameter space ($5 \times 4 \times 4 = 80$ points) provides a comprehensive dataset for training surrogate models or constructing response surfaces.

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